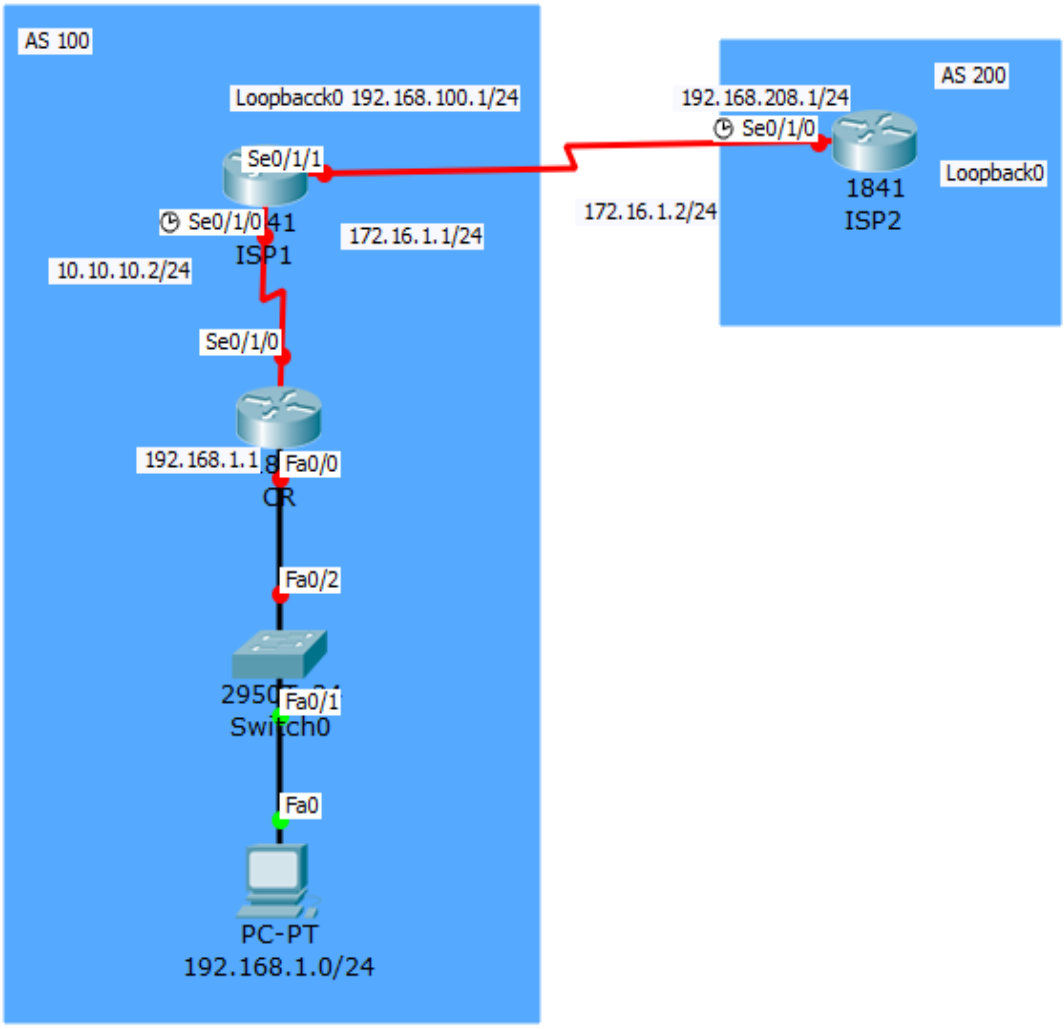
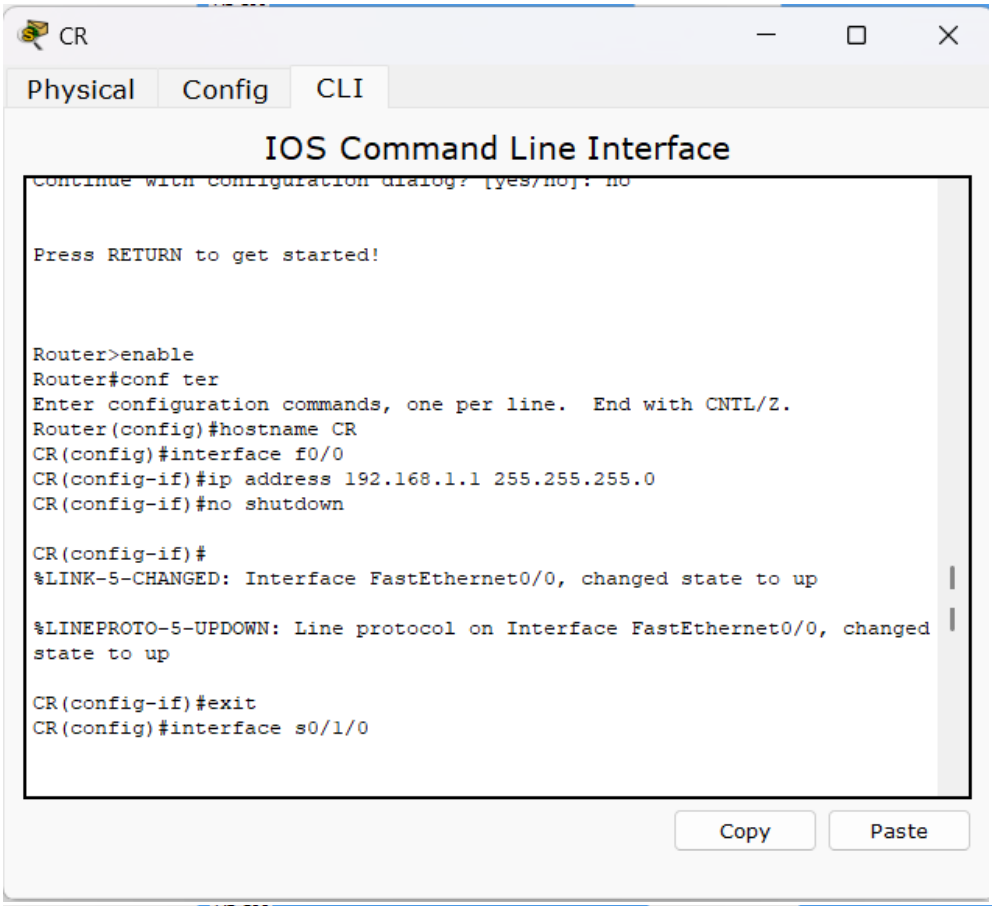
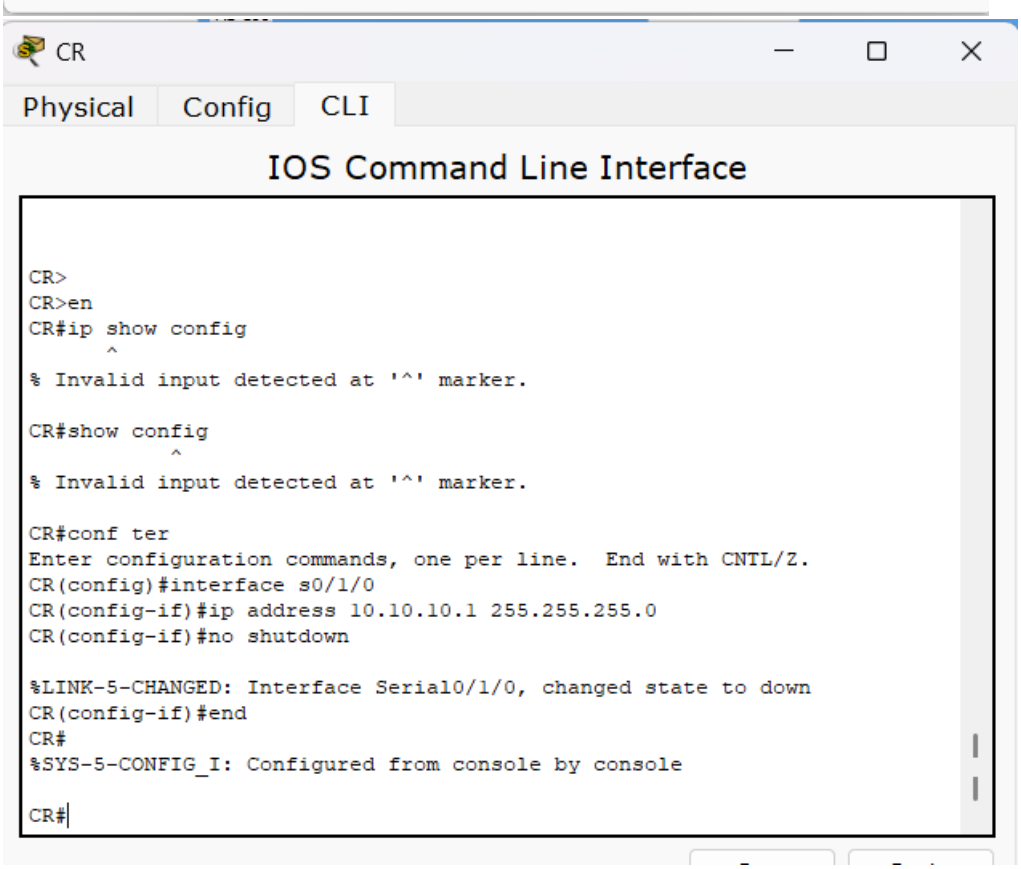
Compte-rendu TP Lab-BGP

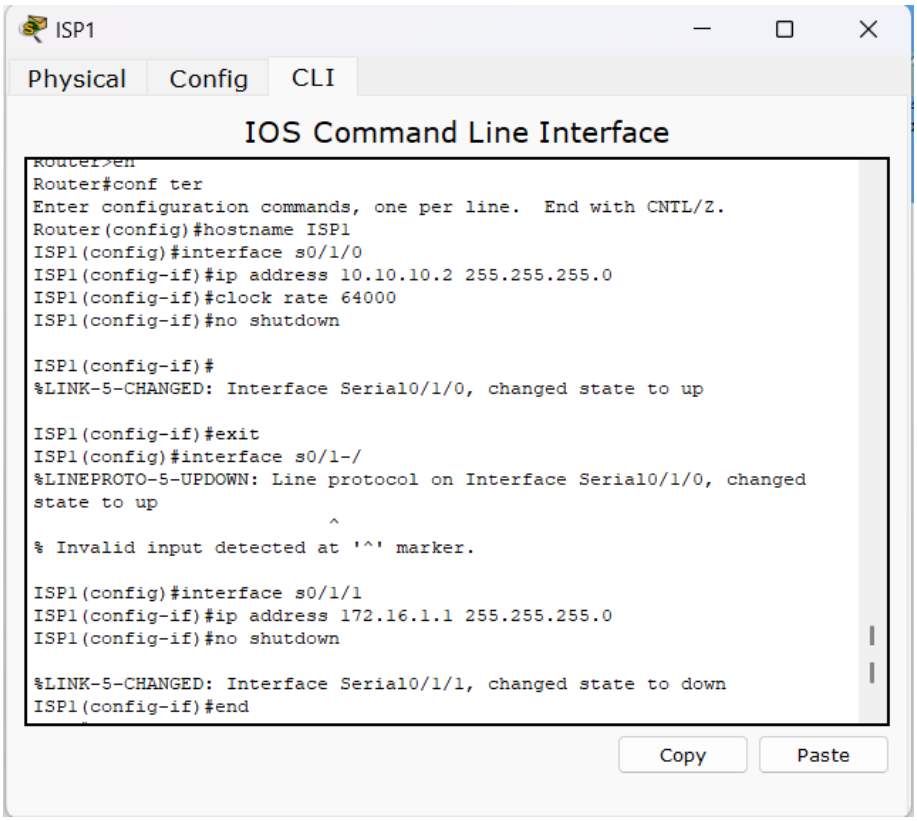
Malek Boughdiri RSI31

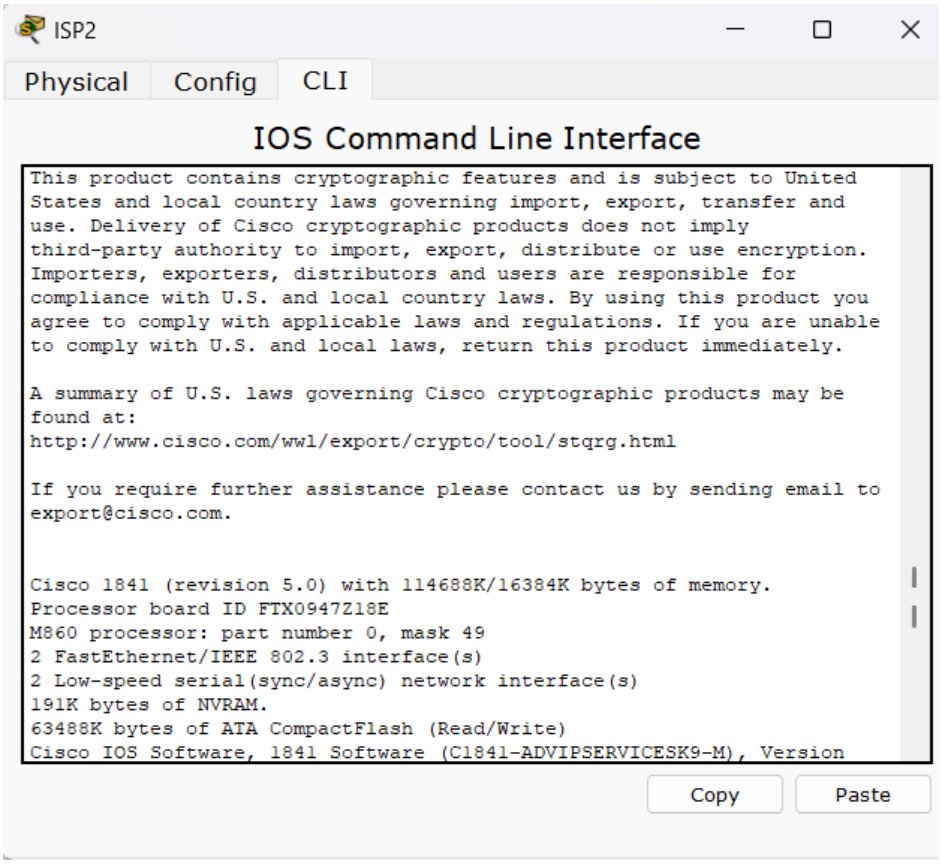


Étape 1 : configuration des informations de base






ISP1
Physical
Config
CLI
IOS Command Line Interface
Router>en
Router#conf ter
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname ISP1
ISP1(config)#interface s0/1/0
ISP1(config-if)#ip address 10.10.10.2 255.255.255.0
ISP1(config-if)#clock rate 64000
ISP1(config-if)#no shutdown
ISP1(config-if)#
%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up
ISP1(config-if)#exit
ISP1(config)#interface s0/1-/
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up
^
% Invalid input detected at '^' marker.
ISP1(config)#interface s0/1/1
ISP1(config-if)#ip address 172.16.1.1 255.255.255.0
ISP1(config-if)#no shutdown
%LINK-5-CHANGED: Interface Serial0/1/1, changed state to down
ISP1(config-if)#end
Copy
Paste


ISP2
Physical
Config
CLI
IOS Command Line Interface
This product contains cryptographic features and is subject to United States and local country laws governing import, export, transfer and use. Delivery of Cisco cryptographic products does not imply third-party authority to import, export, distribute or use encryption. Importers, exporters, distributors and users are responsible for compliance with U.S. and local country laws. By using this product you agree to comply with applicable laws and regulations. If you are unable to comply with U.S. and local laws, return this product immediately.
A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html
If you require further assistance please contact us by sending email to export@cisco.com.
Cisco 1841 (revision 5.0) with 114688K/16384K bytes of memory.
Processor board ID FTX0947Z18E
M860 processor: part number 0, mask 49
2 FastEthernet/IEEE 802.3 interface(s)
2 Low-speed serial(sync/async) network interface(s)
191K bytes of NVRAM.
63488K bytes of ATA CompactFlash (Read/Write)
Cisco IOS Software, 1841 Software (C1841-ADVIPSERVICESK9-M), Version
Copy
Paste

Tester la connectivité :

```
CR>
CR>en
CR#ping 10.10.10.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 3/4/9 ms

CR#
```

```
ISP1#
ISP1#ping 172.16.1.2

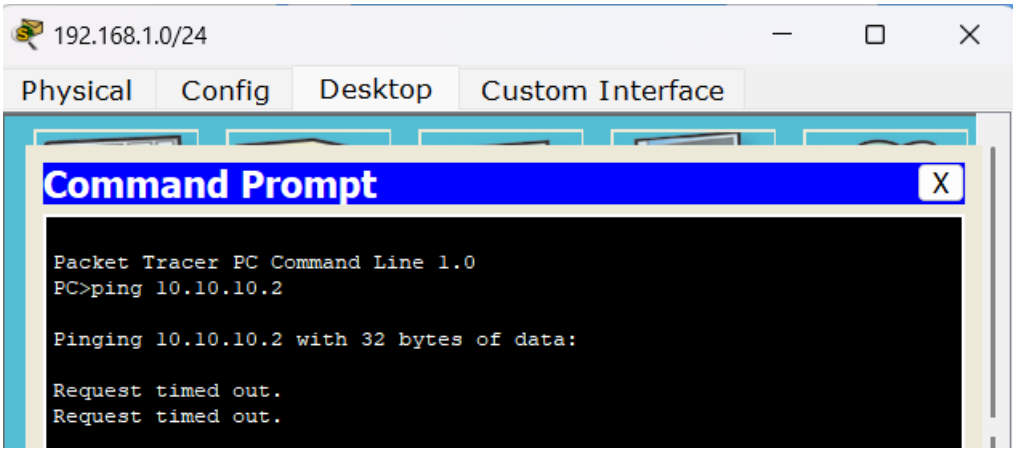
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.1.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 3/5/7 ms

ISP1#
```

```
CR#ping 172.16.1.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.1.2, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

CR#
```



Interface de bouclage sur ISP1 :

```
ISP1(config-if)#
%LINK-5-CHANGED: Interface Loopback0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, chang
to up

ISP1(config-if)#ip address 192.168.100.1 255.255.255.0
ISP1(config-if)#end
ISP1#
%SYS-5-CONFIG_I: Configured from console by console

ISP1#
```

Interface de bouclage sur ISP2 :

```
ISP2(config-if)#
%LINK-5-CHANGED: Interface Loopback0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state
to up

ISP2(config-if)#ip address192.168.200.1 255.255.255.0
                          ^
% Invalid input detected at '^' marker.

ISP2(config-if)#ip address 192.168.200.1 255.255.255.0
ISP2(config-if)#end
ISP2#
%SYS-5-CONFIG_I: Configured from console by console

ISP2#
```

Étape 2 : routes par défaut et routes statiques

```
CR>
CR>en
CR#conf ter
Enter configuration commands, one per line.  End with CNTL/Z.
CR(config)#ip route 0.0.0.0 0.0.0.0 10.10.10.2
CR(config)#
```

```
ISP1#conf ter
Enter configuration commands, one per line.  End with CNTL/Z.
ISP1(config)#ip route 192.168.1.0 255.255.255.0 10.10.10.1
ISP1(config)#
```

Test :

```
PC>ping 10.10.10.2

Pinging 10.10.10.2 with 32 bytes of data:

Reply from 10.10.10.2: bytes=32 time=7ms TTL=254
Reply from 10.10.10.2: bytes=32 time=7ms TTL=254
Reply from 10.10.10.2: bytes=32 time=4ms TTL=254
Reply from 10.10.10.2: bytes=32 time=1ms TTL=254

Ping statistics for 10.10.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 7ms, Average = 4ms
```

Étape 3 : configuration de BGP sur les deux routeurs FAI

```
Enter configuration commands, one per line.  End with CNTL/Z.
ISP1(config)#ip route 192.168.1.0 255.255.255.0 10.10.10.1
ISP1(config)#router bgp 200
ISP1(config-router)#neighbor 172.16.1.1 remote-as 100
ISP1(config-router)#
% Cannot configure the local system as neighbor


ISP1(config-router)#network 192.168.200.0 mask 255.255.255.0
ISP1(config-router)#end
ISP1#
%SYS-5-CONFIG_I: Configured from console by console

ISP1#write memory
Building configuration...
[OK]
ISP1#
```

```
ISP2#conf ter
Enter configuration commands, one per line.  End with CNTL/Z.
ISP2(config)#router bgp 200
ISP2(config-router)#neighbor 172.16.1.1 remote-as 100
ISP2(config-router)#network 192.168.200.0 mask 255.255.255.0
ISP2(config-router)#end
ISP2#
%SYS-5-CONFIG_I: Configured from console by console

ISP2#write memory
Building configuration...
[OK]
ISP2#
```

Étape 4 : affichage des tables de routage

```
ISP2#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS
inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.16.0.0/24 is subnetted, 1 subnets
C       172.16.1.0 is directly connected, Serial0/1/0
B    192.168.1.0/24 [20/0] via 172.16.1.1, 00:33:56
B    192.168.100.0/24 [20/0] via 172.16.1.1, 00:33:56
C    192.168.200.0/24 is directly connected, Loopback0
ISP2#
```

✍ **1) Le réseau 192.168.1.0 est-il dans la table de routage d'ISP2 ? Oui.**

✍ **2) Quelle lettre se trouve à gauche de l'entrée 192.168.1.0 ? B.**

✍ **3) Que signifie cette lettre ? La route a été apprise par le protocole BGP.**

✍ **4) Le réseau 192.168.100.0 est-il dans la table de routage ? Oui.**

✍ **5) Quel routeur a annoncé le réseau 192.168.1.0 ? ISP1 (AS 100), via le voisin 172.16.1.1.**

```
ISP1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS
inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/24 is subnetted, 1 subnets
C       10.10.10.0 is directly connected, Serial0/1/0
     172.16.0.0/24 is subnetted, 1 subnets
C       172.16.1.0 is directly connected, Serial0/1/1
S    192.168.1.0/24 [1/0] via 10.10.10.1
C    192.168.100.0/24 is directly connected, Loopback0
B    192.168.200.0/24 [20/0] via 172.16.1.2, 00:35:48
ISP1#
```

1) Quel(s) réseau(x) sont signalés par ISP2 à ISP1 ? 192.168.200.0/24.

✍ **2) Comment le réseau 192.168.1.0 a-t-il été signalé à ISP1 ? Il n'a pas été signalé par BGP. ISP1 le connaît par une route statique (S) configurée vers CR (10.10.10.1). ISP1 l'annonce ensuite à ISP2 grâce à la commande network.**

✍ **3) ISP1 va-t-il annoncer des réseaux au routeur client ? Non.**

```
CR#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS
inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.10.10.2 to network 0.0.0.0

     10.0.0.0/24 is subnetted, 1 subnets
C       10.10.10.0 is directly connected, Serial0/1/0
C    192.168.1.0/24 is directly connected, FastEthernet0/0
S*   0.0.0.0/0 [1/0] via 10.10.10.2
CR#
```

1) Pourquoi 192.168.100.0 et 192.168.200.0 n'apparaissent-ils pas dans la table de CR ?
Parce que CR n'exécute pas BGP et n'a aucune session avec ISP1. Il n'a qu'une route par défaut statique (S*) vers ISP1, qui lui suffit pour joindre n'importe quel réseau.

Étape 5 : vérification de la connectivité

Ping du PC vers la loopback d'ISP2

```
PC>ping 192.168.200.1

Pinging 192.168.200.1 with 32 bytes of data:

Reply from 192.168.200.1: bytes=32 time=2ms TTL=253
Reply from 192.168.200.1: bytes=32 time=10ms TTL=253
Reply from 192.168.200.1: bytes=32 time=13ms TTL=253
```

Ping d'ISP2 vers le PC

```
ISP2#ping 192.168.1.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.10, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/7/18 ms

ISP2#
```

Étape 6 : affichage des informations BGP

```
ISP1#show ip bgp
BGP table version is 4, local router ID is 192.168.100.1
Status codes: s suppressed, d damped, h history, * valid, > best, i -
internal,
              r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            Metric LocPrf Weight Path
*> 192.168.1.0/24    0.0.0.0                  0      0      0 100 i
*> 192.168.100.0/24  0.0.0.0                  0      0 32768 i
*> 192.168.200.0/24  172.16.1.2               0      0      0 200 i
```

```
ISP2#show ip bgp
BGP table version is 4, local router ID is 192.168.200.1
Status codes: s suppressed, d damped, h history, * valid, > best, i -
internal,
              r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            Metric LocPrf Weight Path
*> 192.168.1.0/24    172.16.1.1               0      0      0 100 i
*> 192.168.100.0/24  172.16.1.1               0      0      0 100 i
*> 192.168.200.0/24  0.0.0.0                  0      0 32768 i
```

Étape 7 : remarques générales

✍ **Pourquoi le routeur ISP1 n'annonce-t-il pas de réseau au routeur client ?**

Parce qu'aucune session BGP n'est configurée entre ISP1 et CR. CR n'a pas de router bgp, et ISP1 n'a pas de commande neighbor pointant vers CR. Le client se connecte à son FAI uniquement par une route par défaut statique.